

Influencer Profitability Prediction for Enhanced Sales Performance

Purpose

To develop a predictive framework that identifies and ranks influencers based on their potential profitability. This framework aims to optimize influencer selection to maximize sales performance and guide strategic decisions for future influencer collaborations.

$$\text{Maximize Influencer Impact} = \frac{\text{Lifetime Influencer Value}}{\text{Lifetime Acquisition Cost}} * \# \text{ Influencers}$$

Prospect Model: This model would focus on initial characteristics without time-series data (i.e., Day 0) to predict short-term potential ILV and engagement. We will use profile-based features (e.g., followers, engagement rate) and platform-related characteristics, which serve as initial proxies for ILV in the absence of longitudinal data.

Ambassador Model: This time-series model can be more robust by incorporating historical data. Using data from Days 0–60 to predict Days 60–90, we will leverage sequential features, such as prior engagement trends and conversion rates, giving a more accurate prediction of future ILV.

Objectives

1. Rank Order Influencers Based on Influencer Lifetime Value (ILV)

- **Goal:** Predict the ILV over the next 30 days using data from the initial 90 days of influencer activity.
- **Approach:** Utilize a combined scoring method that integrates both regression and classification models to provide a more accurate and reliable ranking of influencers.
- **Outcome:** Provide a prioritized list of influencers who are likely to generate the highest profitability, while mitigating overestimation of zero-sales influencers.

2. Identify Key Factors Contributing to Influencer Productivity

- **Goal:** Analyze which attributes of influencers significantly impact sales generation and conversion rates.
- **Approach:** Examine feature importances from both regression and classification models to understand their influence on ILV.
- **Outcome:** Understand the underlying logic behind features that contribute to higher ILV and sales likelihood.

3. Provide Actionable Insights to Improve Productivity

- **Goal:** Offer recommendations on strategies to enhance influencer performance based on model insights.
 - **Approach:** Use model findings to suggest data-driven approaches for optimizing future influencer campaigns.
 - **Outcome:** Deliver strategic guidance to improve influencer engagement and sales outcomes.
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Data Sources and Key Columns

To achieve the objectives, we will utilize data from various sources, focusing on essential columns for analysis.

a. ATTRIBUTIONS Table

- **Key Columns:** `id`, `attribution_mode`, `attributed_total`, `commission`, `influencer_id`, `conversion_id`
- **Usage:** Primary table for linking influencers to conversions and calculating attribution metrics.

b. CONVERSIONS Table

- **Key Columns:** `id`, `current_total_price`, `currency_code`, `platform_order_id`, `is_new_customer`
- **Usage:** Provides details on each conversion, including monetary value and customer type.

c. LINE_ITEMS Table

- **Key Columns:** `id`, `unit_price`, `current_quantity`, `discounted_unit_price`, `conversion_ref`
- **Usage:** Offers product-level details for each conversion, useful for product performance analysis.

d. URL_CLICKS Table

- **Key Columns:** `id`, `created_at`, `referral_link`, `click_type`, `token`, `influencer_id`
- **Usage:** Tracks click activities, essential for measuring influencer engagement.

e. URL_LOOKUP Table

- **Key Columns:** `id`, `token`, `influencer_id`, `clicks`
- **Usage:** Provides aggregated click data for each influencer's unique links.

f. clientshopcustomers Table

- **Key Columns:** `id`, `platformid`, `amountspent`, `currencycode`, `clientid`
- **Usage:** Offers customer-level data for calculating customer lifetime value.

g. prospects Table

- **Key Columns:** `id`, `engagements`, `name`, `email`, `followers`, `avgview`, `socialhandle`
- **Usage:** Provides data on potential influencers for future collaborations.

h. INFLUENCER_STATS Table

- **Key Columns:** `influencer_id`, `total_followers`, `total_engagements`, [Platform-specific metrics]
 - **Usage:** Offers comprehensive statistics on influencer performance across platforms.
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Methodology and Specific Tasks

The project will follow a structured approach, encompassing data preparation, analysis, modeling, and recommendation phases.

1. Data Preparation and Integration

Objective: Merge and align data from various sources to create a comprehensive dataset for analysis.

Tasks:

- **Data Cleaning:**

- Standardize column names to lowercase for consistency.
- Handle missing values through appropriate imputation methods or removal.
- Convert date columns to `datetime` format for accurate time-based analyses.
- **Data Merging:**
 - Merge datasets on common keys (e.g., `influencer_id`, `conversion_id`).
 - Ensure time frames are correctly aligned for the initial 90-day period.
- **Data Quality Assurance:**
 - Remove duplicates and handle data inconsistencies.
 - Flag incomplete records for further review and cautious interpretation.

2. Exploratory Data Analysis (EDA)

Objective: Gain insights into data distributions, relationships, and potential anomalies.

Tasks:

- **Descriptive Statistics:**
 - Calculate mean, median, standard deviation for key metrics (e.g., `total_followers`, `current_total_price`).
- **Data Visualization:**
 - Plot time series of sales and engagements over the 90-day period.
 - Visualize distributions of influencer metrics (e.g., followers, engagement rates).
- **Correlation Analysis:**
 - Compute correlation matrices to identify relationships between variables.
 - Use scatter plots and heatmaps to explore relationships between influencer attributes and sales.

3. Feature Engineering

Objective: Create relevant features that could impact ILV and sales likelihood.

Tasks:

- **Calculate Engagement Metrics:**
 - `engagement_rate = total_engagements / total_followers`
 - `average_engagement_per_post = total_engagements / total_posts`
- **Calculate Sales Metrics:**
 - `average_order_value = total_sales_amount / number_of_sales`
 - `conversion_rate = number_of_conversions / total_clicks`
 - `new_customer_rate = number_of_new_customers / number_of_customers`
- **Time-Based Features:**
 - `days_since_last_post`

- `posting_frequency = total_posts / number_of_days_active`
- **Normalize and Transform Features:**
 - Apply log transformation to skewed features (e.g., `total_followers`, `total_engagements`).
 - Standardize features using `StandardScaler` for modeling.
- **Handle Zero-Inflated Data:**
 - Identify and appropriately transform features with excess zeros.
 - Consider adding small constants before log transformations to handle zeros.

4. Modeling

Objective: Predict ILV over the next 30 days using data from the initial 90 days, utilizing both regression and classification models for a combined scoring approach.

4.1 Model Selection and Rationale

- **Regression Model (CatBoost or LightGBM):**
 - **Purpose:** Predict the continuous ILV for each influencer.
 - **Strengths:**
 - Provides interpretable rankings based on expected future sales.
 - Captures complex non-linear relationships between features and ILV.
 - **Weaknesses:**
 - Tends to overestimate ILV for influencers who will generate zero sales.
 - May not handle zero-inflated data effectively without adjustments.
- **Classification Model (XGBoost):**
 - **Purpose:** Predict the probability of an influencer generating any sales (sales > 0).
 - **Strengths:**
 - Effective in handling class imbalance with proper techniques.
 - Helps secure the "lower bound" by identifying influencers unlikely to generate sales.
 - **Weaknesses:**
 - Risk of false negatives (missing influencers who will generate sales).
 - Provides probabilistic outcomes without estimating sales amounts.

4.2 Combined Scoring Approach

Methodology:

- **Compute Regression Predictions:**
 - Use the regression model to predict ILV ($\hat{Y}_{reg}$) for each influencer.
 - Apply necessary transformations to ensure predictions are in the appropriate scale.
- **Compute Classification Probabilities:**

- Use the classification model to predict the probability (P_{class}) that an influencer will generate sales.
- **Calculate Composite Score:**
 - **Formula:** $\text{Composite Score} = \hat{Y}_{\text{reg}} \times P_{\text{class}}$
 - **Rationale:**
 - Integrates both the expected sales amount and the likelihood of generating sales.
 - Mitigates overestimation by reducing the influence of low-probability influencers.
- **Ranking:**
 - Rank influencers based on the composite score in descending order.
 - Provides a prioritized list that balances potential sales and sales likelihood.

4.3 Model Fine-Tuning

Regression Model Fine-Tuning:

- **Handle Zero-Inflated Data:**
 - Explore models suitable for zero-inflated data, such as Tweedie Regression.
 - Adjust loss functions or apply regularization to prevent overestimation of zero-sales influencers.
- **Hyperparameter Optimization:**
 - Use techniques like GridSearchCV or RandomizedSearchCV for hyperparameter tuning.
 - Implement early stopping to prevent overfitting.
- **Feature Selection:**
 - Utilize feature importance metrics to select predictive features.
 - Remove or penalize features contributing to overestimation of zeros.

Classification Model Fine-Tuning:

- **Address Class Imbalance:**
 - Apply resampling techniques (e.g., SMOTE for oversampling minority class).
 - Adjust class weights or use scale_pos_weight parameter in XGBoost.
- **Reduce False Negatives:**
 - Optimize threshold settings to prioritize recall over precision.
 - Incorporate cost-sensitive learning to penalize misclassification of positive cases.
- **Hyperparameter Optimization:**
 - Tune parameters such as learning rate, max_depth, and n_estimators.
 - Use cross-validation to validate model generalization.

4.4 Model Evaluation Metrics

- **Regression Model Evaluation:**

- **Mean Squared Error (MSE):** Measures average squared difference between predicted and actual ILV.
- **Root Mean Squared Error (RMSE):** Square root of MSE for interpretability in original units.
- **R-squared (R^2):** Proportion of variance in ILV explained by the model.
- **Classification Model Evaluation:**
 - **Precision:** Proportion of true positives among predicted positives.
 - **Recall (Sensitivity):** Proportion of true positives correctly identified.
 - **F1 Score:** Harmonic mean of precision and recall.
 - **Area Under ROC Curve (AUC-ROC):** Measures model's ability to distinguish between classes.
- **Combined Model Evaluation:**
 - **Spearman's Rank Correlation Coefficient:** Assesses the correlation between the ranked predicted composite scores and actual ILV rankings.
 - **Normalized Discounted Cumulative Gain (NDCG):** Evaluates the ranking quality, emphasizing higher ranks.

4.5 Cross-Validation and Model Selection

- **K-Fold Cross-Validation:**
 - Use stratified K-fold for classification to maintain class proportions.
 - Use K-fold for regression to assess model performance across folds.
- **Model Selection Criteria:**
 - Choose models that optimize evaluation metrics while generalizing well to unseen data.
 - Balance between model complexity and interpretability.

5. Analysis of Factors Contributing to ILV and Sales Likelihood

Objective: Identify significant predictors of ILV and the likelihood of generating sales.

Tasks:

- **Feature Importance Analysis:**
 - Extract feature importances from tree-based models (e.g., CatBoost, XGBoost).
 - Analyze how each feature contributes to both ILV prediction and sales likelihood.
- **Statistical Testing:**
 - Perform statistical tests (e.g., t-tests, chi-squared tests) to assess feature significance.
 - Examine correlations between features and target variables.
- **Interaction Effects:**
 - Explore interactions between features that may influence ILV and sales likelihood.
 - Incorporate interaction terms into models if they significantly improve performance.

6. Recommendations and Actionable Insights

Objective: Provide strategic recommendations based on analysis findings.

Tasks:

- **Influencer Selection:**
 - Prioritize influencers with high composite scores for future campaigns.
 - Consider both the expected ILV and the likelihood of generating sales.
 - **Performance Improvement Strategies:**
 - Suggest targeted interventions for influencers with high potential but lower probabilities.
 - Recommend content strategies to increase engagement and conversion rates.
 - **Data-Driven Decision Making:**
 - Encourage the use of model insights for personalized influencer management.
 - Advocate for continuous monitoring and adaptation based on performance metrics.
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Implementation Notes

Data Limitations

- **Incomplete Data:**
 - **Challenge:** Missing engagement metrics or sales data can affect model accuracy.
 - **Handling:**
 - Impute missing values using statistical methods (e.g., median imputation for numerical features).
 - Exclude records with critical missing data if they cannot be reliably imputed.
- **Data Consistency:**
 - **Challenge:** Inconsistent units or formats across datasets.
 - **Handling:**
 - Standardize monetary values to a single currency using exchange rates.
 - Align date formats and ensure all timestamps are in the same time zone.

Time Frame Alignment

- **Initial 90-Day Period:**
 - Ensure all features are calculated using data exclusively from the first 90 days.
- **Prediction Window:**

- The target variables (**ILV_next_30_days** and sales occurrence) correspond to the subsequent 30 days.
- **Data Leakage Prevention:**
 - Avoid using data from the prediction window in feature calculations.
 - Carefully split data to prevent overlap between training and testing periods.

Model Validation

- **Overfitting Prevention:**
 - Use regularization techniques and early stopping in models.
 - Monitor training and validation metrics for divergence.
- **Reproducibility:**
 - Set random seeds for all random processes (e.g., data splitting, model training).
 - Document all parameters and configurations used.

Composite Scoring Implementation

- **Scaling and Calibration:**
 - Ensure regression predictions and classification probabilities are on compatible scales.
 - Calibrate classification probabilities using methods like Platt Scaling if necessary.
 - **Threshold Determination:**
 - Experiment with different thresholds for classification probabilities when computing the composite score.
 - Analyze the impact of thresholds on model performance and business objectives.
 - **Interpretation:**
 - Provide clear explanations of how the composite score is derived and its implications.
 - Ensure stakeholders understand the balance between predicted ILV and sales likelihood.
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Conclusion

By integrating both regression and classification models into a combined scoring approach, we aim to leverage the strengths of each model while mitigating their weaknesses. This methodology provides a more accurate and reliable ranking of influencers based on both their potential to generate sales and the likelihood of doing so. Through careful data preparation, feature engineering, model fine-tuning, and evaluation, we strive to deliver actionable insights that drive strategic decision-making and optimize influencer marketing campaigns.